Chapter 3:
The Real Numbers

Definition 3.1. *The* <u>*real numbers*</u> $\mathbb{R}$ *is an ordered set equipped with two binary operations, addition* $(+)$ *and multiplication* $(\times)$ *such that the following statements are true:*

(O1) *for every* $x, y \in \mathbb{R}$ *exactly one of the following is true:*

- $x < y$
- $y < x$
- $x = y$

(O2) $(\forall x, y, z \in \mathbb{R})\ \ [(x < y) \wedge (y < z)] \Rightarrow x < z$

(A1) $(\forall x, y \in \mathbb{R})\ \ x + y \in \mathbb{R}$

(A2) $(\forall x, y \in \mathbb{R})\ \ x + y = y + x$

(A3) $(\forall x, y, z \in \mathbb{R})\ \ (x + y) + z = x + (y + z)$

(A4) $(\exists 0 \in \mathbb{R})\ \ [(\forall x \in \mathbb{R})\ \ x + 0 = x]$

(A5) $(\forall x \in \mathbb{R})\ \ [(\exists y \in \mathbb{R})\ \ x + y = 0]$

(M1) $(\forall x, y \in \mathbb{R})\ \ xy \in \mathbb{R}$

(M2) $(\forall x, y \in \mathbb{R})\ \ xy = yx$

(M3) $(\forall x, y, z \in \mathbb{R})\ \ (xy)z = x(yz)$

(M4) $(\exists 1 \in \mathbb{R} \setminus \{0\})\ [(\forall x \in \mathbb{R})\ \ 1x = x]$

(M5) $(\forall x \in \mathbb{R} \setminus \{0\})\ \ [(\exists y \in \mathbb{R} \setminus \{0\})\ \ xy = 1]$

(D) $(\forall x, y, z \in \mathbb{R})\ \ x(y + z) = xy + xz$

(OA) $(\forall x, y, c \in \mathbb{R})\ \ (x < y) \Rightarrow (x + c < y + c)$

(OM) $(\forall x, y \in \mathbb{R})\ \ [(0 < x) \wedge (0 < y)] \Rightarrow (0 < xy)$

(C) *every non-empty subset of* $\mathbb{R}$ *that is bounded above in* $\mathbb{R}$ *has a least upper bound in* $\mathbb{R}$

The Axioms of the Real Numbers assert that

- $\mathbb{R}$ is an ordered field;
- $\mathbb{R}$ is complete, i.e. every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.

Notice that $\mathbb{Q}$ is an ordered field but is not complete.

It follows from the Axioms that $\mathbb{R}$ contains $\mathbb{Q}$.

We say $\mathbb{Q}$ is a <u>subfield</u> of $\mathbb{R}$ since $\mathbb{Q} \subset \mathbb{R}$ and $\mathbb{Q}$ is an ordered field with the same order $<$ and operations $+$ and $\times$ as $\mathbb{R}$.

In particular, by (A4) and (M4), $\mathbb{R}$ contains 0 and 1. By the Axioms, $1 > 0$.

By successively adding 1 and using (OA), $\mathbb{N} \subset \mathbb{R}$:

Using (A5) and (M5), it follows that $\mathbb{Z} \subset \mathbb{R}$ and $\mathbb{Q} \subset \mathbb{R}$.

One can show that:

- Using the rational numbers $\mathbb{Q}$, one can construct an ordered field **R** which satisfies the Axioms of the Real Numbers.

  (See the Lecture Notes on Dedekind cuts and Cauchy completion.)

- The ordered field that satisfies the Axioms of the Real Numbers is "unique." That is, every ordered fields satisfying the Axioms "looks the same", or is isomorphic.

Theorem 3.8. *Let* $x, y, z \in \mathbb{R}$ *with* $z > 0$. *If* $x < y$, *then* $xz < yz$.

Idea of Proof.

Proof.

Theorem 3.12. *For all* $x, y \in \mathbb{R}$, *if* $x < y$ *then* $(-y) < (-x)$.

Idea of Proof.

Proof.

Theorem. *Let* 0 *be the additive identity in* $\mathbb{R}$ *and* -0 *be its additive inverse in* $\mathbb{R}$. *Then*

$$-0 = 0.$$

Proof. Since 0 is the additive identity and addition is commutative

$$0 + (-0) = (-0) + 0 = (-0).$$

On the other hand, since -0 is the additive inverse of 0

$$0 + (-0) = 0.$$

Therefore, $-0 = 0 + (-0) = 0$.

Theorem 3.13. *For all* $x \in \mathbb{R}$, *if* $x > 0$ *then* $-x < 0$.

Proof. By Theorem 3.12

$$(-x) < (-0).$$

But $-0 = 0$, so in fact

$$(-x) < 0.$$

We often write

$$a < b < c$$

as a shorthand for

$$a < b \quad \text{and} \quad b < c.$$

Theorem 3.14. *Let $x, y \in \mathbb{R}$. If $x < y$, then*

$$x < \frac{x+y}{2} \quad \textit{and} \quad \frac{x+y}{2} < y.$$

Idea of Proof.

Theorem 3.14. *Let* $x, y \in \mathbb{R}$. *If* $x < y$, *then*

$$x < \frac{x+y}{2} \quad \textit{and} \quad \frac{x+y}{2} < y.$$

Proof. Let's first show that if $x < y$ then $x < \frac{x+y}{2}$.

By similar reasoning, if $x < y$ then $\frac{x+y}{2} < y$.

Definition 3.15. Let $a, b \in \mathbb{R}$ with $a < b$.

The <u>closed interval</u> from a to b is the set

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

Definition 3.16. Let $a, b \in \mathbb{R}$ with $a < b$.

The <u>open interval</u> from a to b is the set

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$$

Due to the flexibility of our notation of closed and open intervals, we can represent intervals that are neither closed nor open as follows:

$$
\begin{aligned}
(a, b] &= \{x \in \mathbb{R} \mid a < x \leq b\} \\
[a, b) &= \{x \in \mathbb{R} \mid a \leq x < b\} \\
(a, \infty) &= \{x \in \mathbb{R} \mid a < x\} \\
[a, \infty) &= \{x \in \mathbb{R} \mid a \leq x\} \\
(-\infty, b) &= \{x \in \mathbb{R} \mid x < b\} \\
(-\infty, b] &= \{x \in \mathbb{R} \mid x \leq b\}
\end{aligned}
$$

Definition 3.17. *Let* $x \in \mathbb{R}$. *We define* $|x|$ *so that*

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Definition. *Let* $x, y \in \mathbb{R}$. *The notation* $|x - y|$ *refers to the distance between* x *and* y *on the real line. When* $y = 0$ *we write* $|x|$ *in place of* $|x - 0|$.

Examples:

$$|x - 5| < 2$$

$$|x + 3| < 2$$

$$|x| \leq a$$

Theorem 3.18. *Let $x, a \in \mathbb{R}$ with $a > 0$. We have that*

$$|x| \leq a \quad \text{if and only if} \quad -a \leq x \text{ and } x \leq a$$

Theorem 3.19. (The Triangle Inequality) *For every $x, y \in \mathbb{R}$ we have that*

$$|x + y| \leq |x| + |y|.$$

Theorem. *For every* $x \in \mathbb{R}$ *we have that*

$$x \leq |x|.$$

Proof. By (O1), we can separately consider the cases $x \geq 0$ and $x < 0$.

Case $x \geq 0$

Case $x < 0$

Theorem. *For every* $x \in \mathbb{R}$ *we have that* $-x \leq |x|$.

Theorem 3.19. (The Triangle Inequality) *For every* $x, y \in \mathbb{R}$ *we have that*

$$|x + y| \leq |x| + |y|.$$

Proof. By (O1), we can separately consider the cases $x + y \geq 0$ and $x + y < 0$.

Case $x + y \geq 0$

Case $x + y < 0$

Theorem 3.20. *For every $x, y \in \mathbb{R}, x < y$, there exists a rational number $r \in \mathbb{Q}$ such that*

$$x < r < y.$$

Theorem 3.21. (Archimedean Principle v1) *The set of natural numbers* $\mathbb{N}$ *is unbounded above in* $\mathbb{R}$.

The following statements are equivalent:

Theorem 3.21. (Archimedean Principle v1) *The set of natural numbers* $\mathbb{N}$ *is unbounded above in* $\mathbb{R}$.

Theorem 3.22. (Archimedean Principle v2) *For every* $x \in \mathbb{R}$ *with* $x > 0$ *there exists* $n \in \mathbb{N}$ *such that* $n - 1 \leq x \leq n$.

Theorem 3.23. (Archimedean Principle v3) *If* $x, y \in \mathbb{R}$ *with* $x, y > 0$ *there exists* $n \in \mathbb{N}$ *such that* $y < nx$.

$$\begin{tikzcd}
 & \textbf{Archimedean Principle v1} \arrow[dl, Leftrightarrow] \arrow[dr, Leftrightarrow] & \\
\textbf{Archimedean Principle v2} \arrow[rr, Leftrightarrow] & & \textbf{Archimedean Principle v3}
\end{tikzcd}$$

Theorem 3.22. (Archimedean Principle v2) *For every* $x \in \mathbb{R}$ *with* $x > 0$ *there exists* $n \in \mathbb{N}$ *such that* $n - 1 \leq x \leq n$.

Proof. By Archimedean Principle v1, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. Hence x is not an upper bound of $\mathbb{N}$. Thus the set

$$\{m \in \mathbb{N} \mid m \geq x\}$$

is non-empty. Choose n to be the smallest element of this set.

Theorem 3.23. (Archimedean Principle v3) *If* $x, y \in \mathbb{R}$ *with* $x, y > 0$ *there exists* $n \in \mathbb{N}$ *such that* $y < nx$.

Proof. Set $t = y/x$.

By Archimedean Principle v1, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. Hence t is not an upper bound of $\mathbb{N}$. Therefore, there exists $n \in \mathbb{N}$ such that $t = y/x < n$.

Theorem 3.20. *For every $x, y \in \mathbb{R}, x < y$, there exists a rational number $r \in \mathbb{Q}$ such that*

$$x < r < y.$$

Proof.